Data and script files for the preparation and pheotypic analysis for the main traits and their derived indices for the 2015 salt experiments on the *indica* and *aus* Rice panels reported in Al-Tamimi et al. (2016)

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August 5, 2016

This document describes the data and code files that were used in the data preparation and phenotypic analysis phases for the two experiments reported by Al-Tamimi et al. (2016): experiment 0254 that investigates an *indica* panel and experiment 0255 that investigates an *aus* panel. These files have been provided in Phenotyping_data_and_scripts.zip.

The main objective of the data preparation phase is to produce growth rates, associated Shoot Ion Independent Tolerance (SIIT) indices and transpiration rates and indices for each experiment. The growth rates are calculated from the raw data and from data smoothed using natural cubic splines. They are calculated using the differences between successive imagings divided by the number of days between the imagings. For the Absolute Growth Rates (AGR); the difference between the values of the response at the two times is taken; while for the Relative Growth Rates (RGR), the difference between the logarithms of the values of the response at the two times is taken. Transpiration is calculated by taking water loss for a pot with a plant and adjusting it for the water loss in the nearest evaporation pot (pot without a plant).

In the phenotypic analysis phase, the traits are subject to a mixed model analyses to produce spatially adjusted BLUEs for each line-treatment combination for each trait. The resulting BLUEs are then subject to association analyses, one for each panel and another for the combined single set. The data and scripts for the latter are available in GWAS_data_and_scripts.zip.

1 Files in Phenotyping_data_and_scripts.zip

1.1 Files in the root directory

This file: Phenotyping_readme.pdf

1.2 Files in the directory *data*

Data input files: Files in csv and xlsx formats containing the original data.

- 0254 rawdata.csv and 0255 rawdata.csv — csv files containing the raw data obtained from the Plant Accelerator for the observed time points for the carts with plants.
- 0254 rawdata.xlsx and 0255 rawdata.xlsx — xlsx files containing the raw data obtained from the Plant Accelerator for the observed time points for the carts with plants.
- 0254 evapdata.csv and 0255 evapdata.csv — csv files containing the raw data obtained from the Plant Accelerator for the observed time points for the evaporation carts (carts with soil but no plants).
- 0254 evapdata.xlsx and 0255 evapdata.xlsx — xlsx files containing the raw data obtained from the Plant Accelerator for the observed time points for the evaporation carts (carts with soil but no plants).
- Evap.carts.csv, Evap.carts.xlsx and Evap.carts.rda — a csv, an xlsx and an R data file with information about which evaporation cart is to be used to adjust water loss for each cart with a plant.

Data files containing the data produced: These files are produced as a result of executing the R data preparation scripts in Section 1.3.

1. 00-raw.0254.dat.rda and 00-raw.0255.dat.rda — R data files produced from executing the script Rice2015.raw.v8.r.
2. 01-longi.0254.dat.rda and 01-longi.0255.dat.rda — R data files produced from executing the script Rice2015.longi.v8.r.
3. 02-cart.0254.dat.rda and 02-cart.0255.dat.rda — R data files produced from executing the script Rice2015.cart.v10.r.
4. 03-longi.0254.SIIT.dat.rda and 03-longi.0255.SIIT.dat.rda — R data files produced from executing the script Rice2015.SIIT.v6.r.
5. Rice2015.0254.v8.xlsx, Rice2015.0255.v8.xlsx — Excel files created using the createXLfile function at the end of Rice2015.spin.v10.r.
6. 0254 Cart BLUEs.v8.csv and from Rice2015.0254.v8.xlsx; 0255 Cart BLUEs.v8.csv and 0254 SIIT BLUEs.v8.csv from Rice2015.0255.v8.xlsx.

1.3 Files in the directory *scripts*

The scripts contain code to be executed using the R statistical computing environment (R Development Core Team, 2015).

R scripts for data preparation: The scripts can be run by executing the sequence of functions in Rice2015.spin.v10.r. They use functions in the R package `imageData`.

1. Rice2015.spin.v10.r — the R script containing the sequence of calls to `spin` from the `knitr` package (see further details below) to execute the other R scripts in this list.
2. Rice2015.raw.v8.r – an R script to import the Excel file containing the raw data from the Plant Accelerator.
3. Rice2015.longi.v8.r — an R script to produce longitudinal values (values for each imaging day) for traits measured in the Plant Accelerator.
4. Rice2015.dfcheck.v3.r — an R script to investigate various values of the smoothing parameter `df` to be used in smoothing PSA.
5. Rice2015.plots.v8.r — an R script to produce plots of the longitudinal data and to identify possibly anomalous plants.
6. Rice2015.cart.v10.r — an R script to produce a single value of each traits for each cart in the Plant Accelerator.
7. Rice2015.cart.plots.v7.r — an R script to produce plots involving the traits for which there is a single value only for each cart in the Plant Accelerator.
8. Rice2015.SIIT.v6.r — an R script to produce differences and ratios of the values of the PSA AGR and RGR for each pair of carts in the Plant Accelerator. The ratios of the PSA RGR are the SIIT values.

R scripts for producing the spatially-adjusted BLUEs: The scripts can be run by executing the sequence of functions in Rice2015.Analyses.v5.spin.r. They use functions in the R packages `asreml`, `asremlPlus`, `dae`, `ggplot2`, `QTLRel` and `XLConnect`.

1. Rice2015.Analyses.v5.spin.r — the R script containing the sequence of calls to `spin` from the `knitr` package (see further details below) to execute the other R scripts in this list.
2. Rice2015.cart.grates.fix.v5.r — an R script to carry out the mixed linear model analysis of the AGRs and RGRs based on Lines fixed and so to produce the BLUEs for them.
3. Rice2015.SIIT.fix.v5.r — an R script to carry out the mixed linear model analysis of the SIITs based on Lines fixed and so to produce the BLUEs for them.

2 Data preparation phase

2.1 The process

The data were input into R statistical computing environment (R Development Core Team, 2015) and processed with the aid of a set of functions collected into the R package `imageData` (Brien, 2016c), available at `http://cran.at.r-project.org/web/packages/imageData/index.html`. The processing was carried out by 'spinning' a sequence of R scripts (see *Rice2015.spin.v10.r*). To 'spin' a script is to execute the function `spin` from the `knitr` package (Xie, 2015) with the name of the script supplied to the function. A 'spin' creates a `html` and an `md` file, each of which contains the output from running the script. Other packages used include `XLConnect` (Mirai Solutions GmbH et al., 2015), `ggplot2` (Wickham, 2016), `GGally` (Schloerke et al., 2016), `Hmisc` (Harrel Jr, 2016) and `dae`(Brien, 2016b), all of which can be downloaded from CRAN (`http://www.r-project.org`).

Five sets of data for each experiment, which are available in R data files, were produced in a seven-step process. The five sets are: (i) Raw data, being the edited data obtained from the Plant Accelerator (step 1); (ii) Longitudinal data, being measurements made on each cart over time (steps 2–4); (iii) Cart data, being a single measurement on the plant in a cart (step 6); (iv) Longitudinal SIIT data, being differences or ratios of

measurements over time calculated from the two plants in neighbouring carts that form a main plot (step 7); (v) Cart SIIT data, being the differences or ratios of measurements calculated from a single measurement on each of the two plants in neighbouring carts that form a main plot (step 7).

The seven-step process, used to produce the data sets and which was repeated in several cycles, is:

1. The raw data is imported from the Excel file supplied by the Plant Accelerator staff and data editing performed. Initially minor editing is performed. In particular, Time.after.Planting..d. is recalculated and the PSA data for the imaging that was to be done on day 40 for Experiment 254 is set to missing because of an equipment malfunction. Also, the name 'final.edit.water.loss' was changed to Water.Loss. Plants that have died or otherwise indentified as being faulty are removed. At subsequent cycles further plants may be excluded, using criteria described below. The script for doing this is *Rice2015.raw.v8.r*.

2. The data was reduced to a subset of the imaging variables produced by the Lemna Tec Scanalyzer and covariates and factors added that might be used in the analysis. In 0254, because of a delay in imaging on day 40, the traits for day 40 are removed, but with adjustment of the water traits. For these experiments, the water calculations are made as follows: (i) the Evaporation data from the empty pots is imported; (ii) the Water.Loss, Water.Amount and empty-pot Evaporation had 100 ml added to their day 30 values because this amount of water was added but not recorded on the salting day (day 29); (iii) in experiment 254, the values of Water.Loss, Water.Amount and Evaporation for day 40 are added to those for day 41 and the values for day 40 set to missing (NA).

3. Derived traits, including continuous growth rates i.e. a growth rate for a cart for each time of observation, except the first, and cumulative responses are obtained. Also Transpiration is calculated for a cart for each time of observation by subtracting from the Evaporation for the nearest evaporation cart from Water.Loss for the cart . Then cumulative Transpiration and cumulative Transpiration Use Efficiency is calculated as the Area at an imaging divided by the cumulative Transpiration. Finally, a continuous Transpiration Use Index (TUE) is calculated as the Total growth since the previous imaging divided by the Transpiration since the previous imaging.

4. Splines are fitted to smooth the longitudinal trends in Total Area and the Transpiration rate. The plots of Area.smooth, AGR and RGR, when different df are used in obtaining Area.smooth, are examined. It was decided to use df equal to 4. Continuous growth rates and TUE indices are calculated from the smoothed Total Area and Transpiration rate. Steps 2–4 are achieved using *Rice2015.longi.v8.r*.

5. An exploratory examination of the unsmoothed data is performed by producing longitudinal plots of unsmoothed and smoothed responses (see *Rice2015.plots.v8.r*). An investigation of anomalous plants is also carried out. The rules used to identify anomalous plants to be further investigated for possible exclusion are:

   - the plant has an Area.smooth value in the interval from Day 40 onwards that is less than 100 for Control plants and less than 75 for Salted plants;
   - the plant has an Area.smooth AGR value in the interval from Day 40 onwards that is less than 2.5;
   - the plant has an Area.smooth.RGR value in the interval from Day 40 onwards that is less than 0.08 for Control plants and less than 0.01 for Salted plants.

6. Produce cart data: traits for which there is a single value for each Snapshot.ID.Tag or cart are obtained. This includes data for a single imaging (e.g. the last date of imaging) and growth rates and TUEs for specified intervals. Responses were produced for the following intervals: Days 31–35, Days 35–38, Days 31–38 and Days 38–42. This is done with the script *Rice2015.cart.v10.r*. Also produced using *Rice2015.cart.plots.v7.r* are scatterplots, with correlation coeffcients, illustrating the relationships between values for different intervals of AGRs, RGRs and TUEs.

7. Produce SIIT data: for each main plot, obtain continuous differences between the treatments for the AGR from the smoothed Total area and continuous SIIT indices for the the AGR and RGR from the smoothed Total area. The script for this is *Rice2015.SIIT.v5.r*.

If, in carrying out the process, plants are identified for exclusion, the above seven-step process is repeated with the the newly-identified plants excluded at step 1. The included scripts are those for the final cycle of the process.

In the end, plants were excluded for the following reasons:

1. Plants that were noted to be dying or dead, or were not loaded;
2. Plants suspected of receiving extra water because of roof leakage;
3. Plants that had not received the correct salt treatment;
4. Control plants that markedly outperformed by its associated Salt-treated plant (SIIT for days 31–35 greater than 1.25)

2.2 Plots produced

Traits have been plotted over time for each plant in longitudinal or profile plots, each of which contains four graphs for a trait: one for each combination of Smarthouse and treatment. In each graph, a line represents the values for a single plant. These plots are produced using *Rice2015.plots.v8.r*. For each experiment, the plots are organised as follows:

Raw or unsmoothed data Plots for the following responses are given:

1. PSA (kpixels)
2. PSA AGR (kpixels per day)
3. PSA RGR (per day)
4. Transpiration rate (mL)
5. Total area TUE (kpixels per mL) = PSA / Transpiration rate
6. Cumulative transpiration (mL)
7. Cumulative Transpiration Use Efficiency = PSA / cumulative Transpiration (kpixels per mL)

Smoothed data Plots are given for the PSA (kpixels) and their AGRS and RGRs and for the Transpiration rate and Total area TUE.

Investigation of anomalies for smoothed Total Area Up to seven plots, as well as a listing of anomalous plants, are provided. Anomalous plants are those with (i) a smoothed Total area from Day 40 onwards less than 100 for Control plants and less than 75 for Salted plants, (ii) an AGR from Day 40 onwards less than 2.5 and (iii) an RGR from Day 40 onwards less than 0.08 for Control plants and less than 0.01 for Salted plants. Plots for anomalous plants are omitted if there are no such plants.

1. Plot of Area.smooth (smoothed PSA) with anomalous plants using (i) removed
2. Plot of Area.smooth (smoothed PSA) for only anomalous plants according to (i)
3. Plot of the AGR with anomalous plants using (ii) removed
4. Plot of the AGR for only anomalous plants according to (ii)
5. Plot of the RGR with anomalous plants using (iii) removed
6. Plot of the RGR for only anomalous plants according to (iii)
7. Plot of Total Area (PSA) values with anomalous plants highlighted in purple

3 Phenotypic analysis phase

3.1 Producing the spatially-adjusted BLUEs

The data contained in 02-cart.0254.dat.rda and 02-cart.0255.dat.rda are analysed using the script Rice2015.cart.grates.fix.v5.r to produce the BLUEs stored in sheets in Rice2015.0254.v8.xlsx and Rice2015.0255.v8.xlsx. The data contained in 03-longi.0254.SIIT.dat.rda and 03-longi.0255.SIIT.dat.rda are analysed using the script Rice2015.SIIT.fix.v5.r to produce the BLUEs stored in sheets in Rice2015.0254.v8.xlsx and Rice2015.0255.v8.xlsx. In doing this the following R packages (R Development Core Team, 2015) are used: ASReml-R (Butler et al., 2009), asremlPlus (Brien, 2016a), dae (Brien, 2016b), ggplot2 (Wickham, 2016), QTLRel (Cheng, 2015) and XLConnect (Mirai Solutions GmbH et al., 2015).

References

Al-Tamimi, N., C. J. Brien, H. Oakey, B. Berger, S. Saade, Y. S. Ho, S. M. Schmöckel, M. Tester, and S. Negrão (2016) Salinity tolerance loci revealed in rice using high-throughput non-invasive phenotyping. *Nature Communications*,. accepted for publication.

Brien, C. J. (2016a) *asremlPlus: augments the use of asreml in fitting mixed models*. URL: http://cran.at.r-project.org/web/packages/asremlPlus/index.html, (R package version 2.0-9, accessed May 17, 2016).

Brien, C. J. (2016b) *dae: functions useful in the design and ANOVA of experiments*. URL: http://cran.at.r-project.org/web/packages/dae/index.html, (R package version 2.7-16, accessed May 17, 2016).

Brien, C. J. (2016c) *imageData: aids in processing and plotting data from a Lemna-Tec Scananalyzer*. URL: http://cran.at.r-project.org/web/packages/imageData/index.html, (R package version 0.1-21, accessed May 17, 2016).

Butler, D., B. R. Cullis, A. R. Gilmour, and B. J. Gogel (2009) *Analysis of Mixed Models for S language environments: ASReml-R reference manual.*. Brisbane: DPI Publications.

Cheng, R. (2015) *QTLRel: Tools for Mapping of Quantitative Traits of Genetically Related Individuals and Calculating Identity Coefficients from Pedigrees*. URL: http://cran.at.r-project.org/web/packages/QTLRel/index.html, (R package version 0.2-15, accessed May 12, 2016).

Harrel Jr, F. E. (2016) *Hmisc: Harrel miscellaneous*. URL: http://cran.at.r-project.org/web/packages/Hmisc/index.html, (R package version 3.17-4, accessed August 3, 2016).

Mirai Solutions GmbH, M. Studer, The Apache Software Foundation, and S. Colebourne (2015) *XLConnect: Excel connector for R*. URL: http://cran.at.r-project.org/web/packages/XLConnect/index.html, (R package version 0.2-11, accessed May 12, 2016).

R Development Core Team (2015) *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. URL: http://www.r-project.org.

Schloerke, B. et al. (2016) *GGally: Extension to 'ggplot2'*. URL: http://cran.at.r-project.org/web/packages/GGaly/index.html, (R package version 1.2.01, accessed August 3, 2016).

Wickham, H. (2016) *ggplot2: An Implementation of the Grammar of Graphics*. URL: http://cran.at.r-project.org/web/packages/ggplot2/index.html, (R package version 2.1.0, accessed May 12, 2016).

Xie, Y. (2015) *knitr: A General-Purpose Package for Dynamic Report Generation in R*. R package version 1.11. URL: http://yihui.name/knitr/.